

UBT620: PROCESS LAB

L T P Cr

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Course objective: The objective of the course is to apply fundamental principles and concepts of bioprocess engineering in bioreactor selection along with its role in manufacturing, processing, and quality control of bioproducts, food, and pharmaceuticals.

Detail contents:

Bioprocess Engineering: Study of thermal death kinetics of microbe, Production of citric acid and lactic acid, comparative study on rate of product formation using immobilized and suspended cells, determination of oxygen uptake rate (OUR) under static and agitated conditions, comparative assessment of cell disruption methods, optimization of flocculating agent concentration, estimation of filtration efficiency of different filter membranes, assessment of adsorption kinetics of phosphate, salting out method for protein precipitation, quantitative and qualitative estimation of products using GC-MS and HPLC, Determination of enzyme kinetics parameters K_m and V_{max}

Pharmaceutical Technology: Quality control of antibiotic and non-antibiotic formulations using titrimetric, spectrophotometric and chromatographic methods as per IP/US Pharmacopoeia, Microbiological assay of the given test antibiotic by cylindrical plate and turbidimetric method and to determine MIC, Sterility testing of formulations: Water for injection, sodium chloride for injection and eye drops

Course Learning Outcomes (CLO):

Students will be able to

1. Comprehend the kinetics of growth and product formation for microbes using different substrates and operational conditions.
2. Understand various primary isolation, recovery and purification operations in bioprocess industries.
3. Analyze the comparative efficiency of food processing and preservation techniques.
4. Chemical and microbiological assays for determining the quality of pharmaceuticals

Textbooks

Approved in 1st meeting of the Academic Council held on September 11, 2025. Further revision approved in 3rd meeting of the Academic Council held on March 18, 2026, incorporating modification in credits due to the addition of AI courses.

1. Stanbury PF, Hall SJ, and Whitaker A, Principles of Fermentation Technology, 3rd Edition, Elsevier (2016).
2. Shuler ML and Kargi F, Bioprocess Engineering, 2nd Edition Prentice Hall (2015).
3. Sivasankar B, Bioseparations: Principles and Techniques, PHI Learning Pvt. Ltd. (2006).
4. Belter PA, Cussler E and Hu WS, Bioseparation – Downstream Processing for Biotechnology, Wiley Interscience (1988).

Reference Books

1. Arvind N. Shukla, Laboratory Bioprocess Technology, Discovery Publishing House (2013).
2. Sharma SK, Mulvaney SJ, Rizvi SHS, Food Process Engineering: Theory and Laboratory Experiments, John Wiley & Sons Inc; 1st edition (30 November 1999); CBS Publishers & Distributors Pvt. Ltd.
3. Indian Pharmacopoeia

Evaluation Scheme:

Sr. No.	Evaluation Elements	Weightage (%)
1.	Sessionals (May include assignments / quizzes / performance)	100

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